Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning

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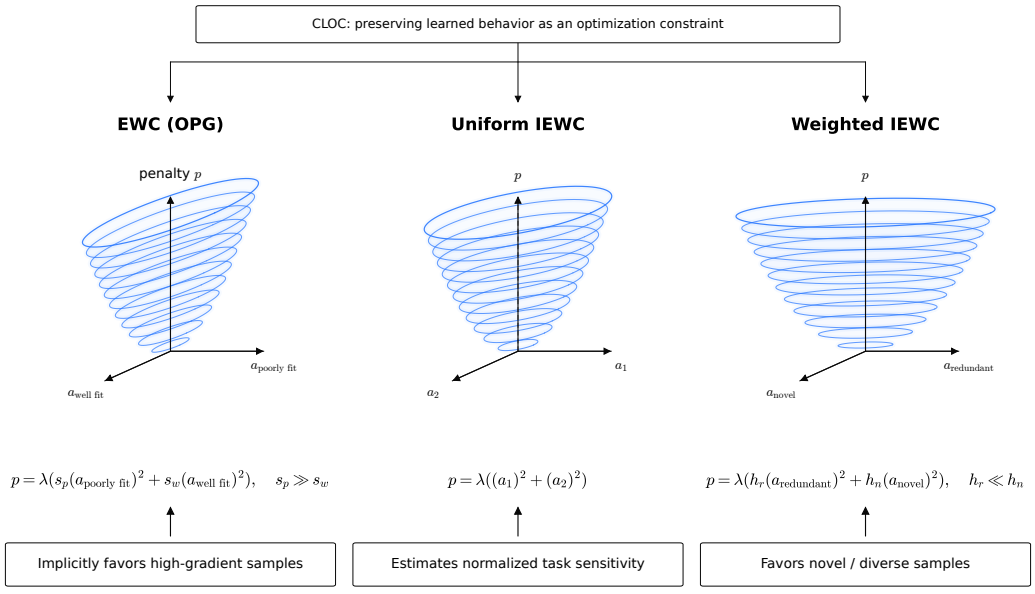


Figure 1: Each panel shows the continual-learning penalty over normalized directions for two representative samples. EWC (OPG) inherits sample weighting from output-loss-gradient scale; uniform IEWC removes this scale away from the damping regime, while residual-diversity weighting changes the aggregation measure as a separate design choice.

Abstract

Elastic Weight Consolidation (EWC) is a widely used regularization-based technique for continual learning, protecting performance on old tasks with a quadratic penalty whose empirical Fisher is usually justified through probabilistic curvature. We instead introduce *Continual Learning as an Optimization Constraint* (CLOC): new training must preserve chosen functionals of learned behavior. EWC's empirical Fisher is exactly the first-order form of this constraint when the preserved functional is each sample's loss. This derivation exposes a flaw: EWC implicitly weights each sample by the scale of its loss gradient, so well-fitted samples lose importance. EWC-DR observed this failure in classification; CLOC identifies its general cause. Improved Elastic Weight Consolidation (IEWC) penalizes output changes according to the first-order approximation of each old sample's loss level set, removing the weighting for any differentiable loss and output geometry, and prevents corrupt samples from dominating the importance matrix. Across classification, segmentation, language, forecasting, and diffusion, IEWC variants match or improve on EWC and outperform EWC-DR. In a held-out evaluation with validation-selected hyperparameters, uniform IEWC improves ImageNet-R

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accuracy over matched EWC by 2.00 percentage points (95% CI 0.93–3.07).

1 CONTINUAL LEARNING AS AN OPTIMIZATION CONSTRAINT

Continual learning asks a model to learn from sequentially available distributions without unrestricted access to earlier data (McCloskey and Cohen, 1989; French, 1999; Parisi et al., 2019; van de Ven and Tolias, 2019). After training on distribution A and obtaining $\theta_A^\star$, minimizing only the later loss $L_B(\theta)$ leaves previously learned behavior unconstrained. Methods such as EWC, SI, MAS, and LwF address this problem with different preservation objects: parameter importance, learning-path importance, output sensitivity, or old predictions (Kirkpatrick et al., 2017; Zenke et al., 2017; Aljundi et al., 2018; Li and Hoiem, 2016). This variety raises a prior question: *what quantity should a local continual-learning penalty preserve?*

We pose that question directly as constrained optimization. Let $q : \Theta \to \mathcal{H}$ be a quantity whose old value or acceptable set represents preservation, and let $\mathcal{C} \subset \mathcal{H}$ encode the corresponding constraint. Learning distribution B becomes

$$\min_{\theta \in \Theta} L_B(\theta) \qquad \text{subject to} \qquad q(\theta) \in \mathcal{C}.$$

We call this perspective *Continual Learning as an Optimization Constraint* (CLOC). CLOC is deliberately a formulation rather than a single algorithm: different choices of q, $\mathcal{C}$, relaxation, and approximation recover different local preservation penalties.

The paper develops one path through this design space. First, a population and low-order relaxation turns samplewise preservation objects into a stored quadratic penalty. Second, preserving any differentiable scalar functional yields a gradient outer-product matrix; choosing the old-sample loss gives OPG, and negative log likelihood gives the empirical Fisher used in common EWC implementations. Third, preserving output motion only in the normal quotient of the old loss level set yields the iEF normalization and IEWC. We make damping, sample weighting, output metric, and matrix approximation explicit design choices instead of treating them as consequences of the normalization.

Concretely, the paper contributes (i) a functional-preservation construction whose loss specialization derives empirical-Fisher EWC and level-set-normalized IEWC within one framework; (ii) a matched output-metric construction and the exact finite-damping, interpolation, and loss-unit boundary; and (iii) a direct spectral test of the mechanism, a matched held-out evaluation with validation-selected hyperparameters, and results across five domains.

Table 1: Core notation; Appendix I gives the complete table.

Symbol	Meaning
J_i, r_i, g_i	Output Jacobian, output-loss gradient, and $g_i = J_i^\top r_i$.
M_{OPG}	Mean $g_i g_i^\top$ from scalar preservation.
$s_i = r_i^\top G_i^{-1} r_i$	Squared dual norm of the output-loss gradient.
M_{IEF}	Mean $g_i g_i^\top/(s_i + \tau_A)$ from level-set preservation.

2 RELATED WORK AND SCOPE

EWC stores a quadratic approximation around earlier parameters and weights parameter motion by a Fisher-derived importance estimate (Kirkpatrick et al., 2017). Other parameter-importance regularizers estimate importance from the optimization trajectory, output sensitivity, or a combination of EWC-style curvature and path-integral evidence (Zenke et al., 2017; Aljundi et al., 2018; Chaudhry et al., 2018); their apparently different estimators have also been analyzed within a common framework (Benzing, 2022). Bayesian and posterior-combination approaches instead accumulate task posteriors or merge their moments (Schwarz et al., 2018; Nguyen et al., 2018; Lee et al., 2017). A practical ambiguity remains: implementations called "Fisher" may use the model-label expectation, sampled model labels, observed labels, or minibatch aggregates. These estimators are not interchangeable in theory or practice (Kunstner et al., 2019; van de Ven, 2025). Our scalar-loss derivation concerns the observed-label OPG object; sampled model-Fisher EWC is evaluated separately rather than treated as another name for the same matrix.

Regularization can instead protect functions or representations. LwF distills old predictions on later inputs, encoder-based lifelong learning constrains reconstructions of old-task features, and functional regularization preserves selected predictive functions or memorable points (Li and Hoiem, 2016; Rannen et al., 2017; Titsias et al., 2020; Pan et al., 2020). More generally, many training objectives contain vectors or matrices that are computed from parameters or activations but held constant with respect to the parameters being optimized, such as weight-averaged teacher parameters (Tarvainen and Valpola, 2017), stop-gradient target projections (Grill et al., 2020; Chen and He, 2021), predictions of an earlier model (Li and Hoiem, 2016), and running activation moments (Wiest, 2026).

EWC and IEWC follow the same pattern in the sense that both compute an importance matrix once earlier tasks are learned and then hold it fixed as the weighting of a quadratic penalty. CLOC complements this taxonomy: it begins with a preservation specification and asks which local stored penalty follows. The algebra below allows differentiable scalar functionals, while IEWC specializes to scalar-loss preservation and loss-level-set-normal output preservation.

iEF was introduced as a normalized observed-gradient approximation for natural-gradient optimization (Wu et al., 2024). We recover its Euclidean algebra from a different question: the metric distance of an old output displacement from the tangent to its old loss level set. This provides a preservation interpretation but does not convert iEF into exact global curvature. Our additions relative to simply substituting iEF into EWC are the CLOC derivation, the matched output-metric generalization, the finite-damping and interpolation boundary, and the separation of normalization from aggregation weights. The RDW ablation borrows a streaming residual-diversity intuition from GSS (Aljundi et al., 2019); it is an explicitly order-dependent heuristic, not part of the level-set result.

3 POPULATION SETTING AND APPROXIMATIONS

For a closed convex set $\mathcal{C}$, the Moreau envelope of its indicator is a scaled squared distance (Parikh and Boyd, 2014). This motivates the soft CLOC objective

$$\begin{aligned}\min_\theta J(\theta) &= \min_\theta \left[L_B(\theta) + \frac{\lambda}{2} \operatorname{dist}(q(\theta), \mathcal{C})^2 \right] \\ &= \min_\theta \left[L_B(\theta) + \frac{\lambda}{2} \inf_{u \in \mathcal{C}} \|q(\theta) - u\|^2 \right].\end{aligned}$$

GEM and A-GEM are similar in spirit in that they formulate preservation through constraints on old examples (Lopez-Paz and Ranzato, 2017; Chaudhry et al., 2019). CLOC turns the old-task constraint into a penalty built after task-A training. In a population setting, we use sample-based analogues ℓ_B and q and take the empirical average. Define the unscaled preservation loss as $p(a, \theta) = \operatorname{dist}(q(a, \theta), \mathcal{C})^2$.

$$\min_\theta \ \frac{1}{N_B} \sum_{i=1}^{N_B} \ell_B(b_i, \theta) + \frac{\lambda}{2N_A} \sum_{i=1}^{N_A} p(a_i, \theta)$$

Additional old distributions enter through analogous constraints. We refer to this population version as **population CLOC (P-CLOC)**. One efficiency consideration is the penalty p: Ideally, we let the constraint contribute to each gradient step with its gradient. However, recomputing the full old-distribution penalty p and its associated factors many times during optimization on distribution B is usually prohibitively expensive, since it depends on the dataset D_A associated with A. Additionally, reusing dataset D_A later on does not respect the continual learning setting. For this reason, replace the exact penalty by a local k-th order Taylor-polynomial approximation around the old solution $\theta_A^\star$. This turns the computation of p into a repeated evaluation of precomputed derivative tensors and bounds memory requirements to $O(d^k)$ for a d-dimensional parameterization. Since memory scales exponentially with k, the standard scalable surrogate is a low-order Taylor approximation. We refer to this local approximation as **low-order CLOC (L-CLOC)**, and to its population version as **PL-CLOC**. It remains to specify the preserved quantity q and constraint set $\mathcal{C}$. Notice that the gradient of p is dependent on only one sample at once, which allows us to analyze p and its gradients on a per-sample basis.

3.1 Scalar Functional Preservation

Let $s_a : \Theta \to \mathbb{R}$ be a differentiable scalar functional of the model on old sample a. Preserving that functional means

$$q(a, \theta) = s_a(\theta) - s_a(\theta_A^\star).$$

Since preservation means $q(a, \theta) = 0$, take $\mathcal{C} = \{0\}$. The corresponding unscaled proximal loss is

$$p(a, \theta) = \operatorname{dist}(q(a, \theta), \mathcal{C})^2 = q(a, \theta)^2.$$

To obtain a low-order Taylor approximation, use the perturbation around the old solution $\delta = \theta - \theta_A^\star$ in the first-order expansion of q:

$$q(a, \theta_A^\star + \delta) \approx g_a^\top \delta,$$

where

$$g_a = \nabla_\theta s_a(\theta)\big|_{\theta = \theta_A^\star}.$$

When the functional factors through the model output as $s_a(\theta) = \phi_a(f_\theta(x))$, write

$$\begin{aligned} z_{A,i} &= f_{\theta_A^\star}(x_i), \\ r_i &= \nabla_z \phi_{a_i}(z)\big|_{z = z_{A,i}}, \\ J_i &= \nabla_\theta f_{\theta_A^\star}(x_i), \end{aligned}$$

the chain rule gives

$$g_i = J_i^\top r_i.$$

Substituting this expansion into the preservation penalty gives

$$\frac{\lambda}{2N_A} \sum_{i=1}^{N_A} q(a_i, \theta)^2 \ \approx \ \frac{\lambda}{2N_A} \sum_{i=1}^{N_A} (g_i^\top \delta)^2.$$

Collecting the gradient outer products yields

$$\frac{\lambda}{2N_A}\sum_{i=1}^{N_A}(g_i^\top\delta)^2=\frac{\lambda}{2}\,\delta^\top\left(\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top\right)\delta=\frac{\lambda}{2}\,\delta^\top M_1\delta.$$

Therefore the resulting matrix is

$$M_1=\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top=\frac{1}{N_A}\sum_{i=1}^{N_A}J_i^\top r_ir_i^\top J_i.$$

The construction is exact for the squared first-order surrogate displayed above, not for the original nonlinear equality constraint. The rest of the paper takes $\phi_a=\ell_A$ unless stated otherwise. This loss specialization is the *outer-product loss-gradient* (OPG) matrix, and for negative log likelihood it becomes the empirical-Fisher algebra used by EWC, as specified next.

3.2 Outer-Product Loss-Gradient Regularization and EWC

For the loss functional $s_{a_i}(\theta)=\ell_A(f_\theta(x_i))$, the matrix obtained in the previous section is

$$M_{\mathrm{OPG}}=\frac{1}{N_A}\sum_{i=1}^{N_A}g_ig_i^\top.$$

It is an outer product of loss gradients. This identity holds for any differentiable loss, without a probabilistic model or a curvature interpretation. We therefore call it an *outer-product-of-loss-gradients* (OPG) matrix.

For a negative-log-likelihood loss,

$$\ell_i(\theta)=-\log p_\theta(y_i\mid x_i),$$

we have $g_i=-\nabla_\theta\log p_\theta(y_i\mid x_i)$. The same OPG algebra then coincides with the observed-label score-gradient matrix commonly called the empirical Fisher,

$$F_{\mathrm{emp}}=\frac{1}{N_A}\sum_{i=1}^{N_A}\nabla_\theta\log p_\theta(y_i\mid x_i)\nabla_\theta\log p_\theta(y_i\mid x_i)^\top.$$

Thus OPG-EWC and empirical-Fisher EWC denote the same implementation in an explicit likelihood setting. The CLOC derivation identifies this matrix as the quadratic form of a squared first-order loss-preservation surrogate, an interpretation distinct from model-Fisher and Hessian curvature.

Writing $s_\theta(x,y)=\nabla_\theta\log p_\theta(y\mid x)$, the model Fisher is instead

$$F_{\mathrm{model}}(\theta)=\mathbb{E}_{x\sim\mathcal{D}_A}\mathbb{E}_{y\sim p_\theta(\cdot|x)}\left[s_\theta(x,y)s_\theta(x,y)^\top\right].$$

It averages over labels sampled from the model, not only the observed label. Under standard regularity conditions this Fisher is connected to expected negative-log-likelihood curvature and natural-gradient geometry (Amari, 1998; Martens, 2020; Martens and Grosse, 2015). The observed-label empirical Fisher is a different object and can be a poor curvature approximation (Kunstner et al., 2019); Fisher-estimator choice can also materially affect EWC behavior and regularization scale (van de Ven, 2025).

Our experiments include a one-label Monte Carlo model-Fisher baseline, obtained by sampling from the frozen predictive distribution at each supervised output site—one class per image and one token per supervised TRACE answer position—alongside OPG-EWC.

4 LEVEL-SET PRESERVATION IN OUTPUT SPACE

Section 3.1 preserves a scalar functional value directly. A second construction instead preserves the model output up to motions along a level set of a differentiable output functional $\phi_a:\mathcal{Z}\to\mathbb{R}$. This output-space perspective connects to distillation and functional regularization approaches to continual learning (Li and Hoiem, 2016; Hinton et al., 2015; Titsias et al., 2020; Pan et al., 2020). Let $a\in D_A$ be an old sample, write $z_A=f_{\theta_A^\star}(x)$, and define the exact preservation set

$$\mathcal{C}_a=\{z:\phi_a(z)=\phi_a(z_A)\}.$$

The preserved quantity and hard constraint are, respectively,

$$q(a,\theta)=f_\theta(x),\qquad q(a,\theta)\in\mathcal{C}_a,$$

and the corresponding unscaled proximal loss is

$$p(a,\theta)=\mathrm{dist}_G\left(q(a,\theta),\mathcal{C}_a\right)^2,$$

where $G\succ 0$ is the chosen local output metric. Positive definiteness makes the primal metric and its inverse on covectors unambiguous; Section 6 treats the restricted pseudoinverse convention needed by a singular ordered-output example.

This exact penalty is generally not tractable. To obtain a practical form, use the standard first-order local model of the constraint set near the old output z_A, namely its tangent plane. Let

$$r=\nabla_z\phi_a(z)\big|_{z=z_A}.$$

We first assume $r\neq 0$, so the functional's level set has a regular tangent hyperplane at z_A. For a small output displacement $\Delta z=f_\theta(x)-f_{\theta_A^\star}(x)$, its first-order change is approximately $r^\top\Delta z$. Hence the local constraint set is

$$T=\{\Delta z:r^\top\Delta z=0\}.$$

Replacing the exact level set by this tangent hyperplane gives the local proximal loss

$$p(a,\theta) \approx \mathrm{dist}_G(\Delta z, T)^2.$$

For the metric $|v|_G^2 = v^\top G v$, the squared distance to the tangent hyperplane follows from projecting Δz onto the gradient:

$$\mathrm{dist}_G(\Delta z, T)^2 = \frac{(r^\top \Delta z)^2}{r^\top G^{-1} r}.$$

Thus G measures primal output displacement while G^{-1} acts on the loss covector; Appendix H.1 gives the projection calculation establishing this matched pair. Linearizing the model output around $\theta_A^\star$, with

$$\delta = \theta - \theta_A^\star, \qquad J = \nabla_\theta f_{\theta_A^\star}(x),$$

we have

$$\Delta z \approx J\delta.$$

Substituting this into the local regularization term gives

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \frac{(r^\top J\delta)^2}{r^\top G^{-1} r}.$$

Equivalently,

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \delta^\top \left(J^\top \frac{r r^\top}{r^\top G^{-1} r} J \right) \delta.$$

Taking the empirical average over old-distribution samples $a_1, \ldots, a_N$ yields an importance matrix of the form

$$M_{2,A} = \frac{1}{N} \sum_{i=1}^{N} J_i^\top \frac{r_i r_i^\top}{r_i^\top G_i^{-1} r_i} J_i.$$

This is a local construction: it penalizes motion normal to the tangent model, not distance to a globally convex neural-network level set. IEWC is the specialization $\phi_a = \ell_A$ studied below; other functionals define CLOC extensions but are not evaluated here. When $r_i = 0$, the tangent normal is undefined and the undamped quotient is $0/0$; our finite-damping implementation assigns that sample zero contribution. Section 5 makes the resulting near-interpolation boundary explicit.

4.1 Level-Set Normalization and the Improved Empirical Fisher

For the loss-functional specialization $\phi_a = \ell_A$ and Euclidean output metric $G_i = I$, the matrix from the previous section is the improved empirical Fisher (iEF) introduced by Wu et al. (Wu et al., 2024):

$$M_{2,I} = F_{\mathrm{IEF}}.$$

Hence iEF is the quadratic form of a first-order local constraint that protects old behavior geometrically: instead of preserving a raw scalar-loss coordinate, it penalizes only output motion normal to the old loss level set. The original iEF paper derives the same normalization for approximate natural-gradient descent. In that setting, the observed-label OPG reweights sample contributions by squared output-gradient magnitude; iEF removes this scale factor and retains the normalized output-loss direction. Here the normalization arises from metric projection onto the normal quotient of the old-loss tangent space.

For scalar least-squares regression in Euclidean geometry, Appendix H.2 shows that undamped IEF is exactly the Gauss–Newton or output-Jacobian matrix (Schraudolph, 2002; Martens, 2010, 2020; Wu et al., 2024). This matches our geometric intuition: loss preservation is governed by how parameters move the output, which is encoded by the Jacobian J, not by the current output-gradient magnitude.

4.2 Normalization and Sample Weighting

To compare both matrices in a common notation, let $G_n \succ 0$ denote the chosen output metric for sample n. Although M_{OPG} itself does not depend on G_n, its summands can be rewritten relative to that metric. In a negative-log-likelihood setting, $M_{\mathrm{OPG}} = F_{\mathrm{emp}}$.

Define

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \quad a_n^G = J_n^\top \hat{r}_n^G, \quad w_n^G = r_n^\top G_n^{-1} r_n.$$

Then, wherever $r_n \neq 0$,

$$M_{\mathrm{IEF},G} = \frac{1}{N} \sum_{n=1}^{N} a_n^G (a_n^G)^\top,$$

$$M_{\mathrm{OPG}} = \frac{1}{N} \sum_{n=1}^{N} w_n^G \, a_n^G (a_n^G)^\top.$$

Both matrices empirically aggregate the same local old-task sensitivity directions $a_n^G (a_n^G)^\top$, but they differ in how strongly each sample contributes. Undamped IEF keeps uniform empirical weighting of these normalized directions, while OPG multiplies each contribution by the metric-dependent squared magnitude

$$w_n^G = r_n^\top G_n^{-1} r_n.$$

Thus normalization changes sample aggregation: OPG shifts old-task weighting toward samples with large output-loss gradients and away from confidently fitted ones, whereas undamped IEF removes that scale.

The next section treats probabilistic classification, where the scale difference has a concrete well-fitted limit.

5 GRADIENT-SCALE BEHAVIOR NEAR WELL-FITTED SOLUTIONS

The output-gradient weighting in observed-label OPG has an important classification symptom. For cross-entropy with softmax probabilities p_n and one-hot label y_n, the output-loss gradient with respect to logits is

$$r_n = p_n - y_n.$$

For a confidently correct prediction with class c and small ε,

$$p_{n,c} = 1 - \varepsilon, \qquad \|r_n\|_2^2 \in \Theta(\varepsilon^2).$$

Thus the OPG contribution becomes small as the model becomes more confident and correct. This is coherent with its derivation: the scalar first-order loss change is also small. For continual-learning preservation, however, small current loss need not mean that losing the sample later is harmless. EWC-DR identifies this classification symptom and proposes a logit-reversal repair (Liu and Chang, 2026).

This observation is one instance of the weighting effect identified in Section 4.2: observed-label OPG importance depends on output-gradient scale. Undamped IEF removes this particular factor. Positive damping retains the removal only above its transition scale, and empirical benefit remains a question for the controlled comparisons below.

The same observation extends to arbitrary differentiable losses.

Importance is estimated after training on A, when many old samples may be well fit. The original EWC Laplace argument likewise expands at an old-task optimum, although its model-Fisher motivation differs from the observed-label OPG analyzed here (Kirkpatrick et al., 2017; Kunstner et al., 2019). Modern expressive models can fit training data up to interpolation (Belkin et al., 2019), so the near-zero regime must be explicit. For one sample, let $s_n = r_n^\top G_n^{-1} r_n$ and

$$S_n^{\mathrm{OPG}} = J_n^\top r_n r_n^\top J_n, \qquad S_{n,\tau}^{\mathrm{IEF},G} = \frac{J_n^\top r_n r_n^\top J_n}{s_n + \tau}.$$

Proposition 1 (Gradient-scale behavior near a well–fitted solution)**.** *Assume $m_G I \preceq G_n \preceq M_G I$ and $\|J_n^\top v\|_2 \le C_J \|v\|_2$. For $s_n > 0$,*

$$S_n^{\mathrm{OPG}} = s_n S_{n,0}^{\mathrm{IEF},G}, \qquad \begin{aligned} \|S_n^{\mathrm{OPG}}\|_2 &\in O(s_n), \\ \|S_{n,0}^{\mathrm{IEF},G}\|_2 &\in O(1). \end{aligned}$$

If $\|J_n^\top r_n/\sqrt{s_n}\|_2 \ge c_J > 0$, these rates are $\Theta(s_n)$ and $\Theta(1)$. Consequently, for Euclidean softmax cross-entropy with correct-class probability $1-\varepsilon$, they are $\Theta(\varepsilon^2)$ and $\Theta(1)$, respectively. For $\tau > 0$,

$$S_{n,\tau}^{\mathrm{IEF},G} = \frac{s_n}{s_n + \tau} S_{n,0}^{\mathrm{IEF},G},$$

so fixed positive damping restores attenuation and $S_{n,\tau}^{\mathrm{IEF},G} \to 0$ as $r_n \to 0$. Finally,

$$\tau_A = \kappa \,\mathrm{median}\{s_i : s_i > 0\}$$

makes every damped summand invariant to a global positive rescaling of the task loss.

Appendix G gives the proof.

The proposition isolates the central EWC–IEWC difference. OPG weights each normalized parameter-sensitivity direction by the squared dual norm s_n of the current output-loss gradient; undamped IEF removes that factor, although model sensitivity through J_n remains. Thus a small current loss gradient need not make the corresponding old behavior unimportant for continual learning.

Finite damping introduces a transition rather than uniformly removing gradient scale: normalization applies when $s_n \gg \tau$, while $s_n \ll \tau$ again suppresses the sample. At exact interpolation we define the contribution as zero; the undamped expression is $0/0$ and has no canonical value without a directional limit. We use $\kappa = 10^{-2}$ in the held-out evaluation. Relative damping is invariant only to global loss-unit changes, not arbitrary sample-dependent transformations; if all recorded s_i vanish, the task importance is the zero matrix.

6 OUTPUT-SPACE GEOMETRY

The scalar CLOC construction and its OPG matrix require no output-space metric. The level-set construction does: for a symmetric positive-definite primal metric G_n, output displacements have inner product $\langle u, v\rangle_{G_n} = u^\top G_n v$, the metric-normal direction is $G_n^{-1} r_n$, and the normal-distance denominator is $r_n^\top G_n^{-1} r_n$. Thus a geometry choice must supply a matched pair: G_n acts on primal output displacement and G_n^{-1} on the loss covector. An arbitrary denominator does not constitute a new geometry.

The canonical choice $G_n = I$ recovers iEF (Wu et al., 2024). More generally, any smooth output discrepancy whose second-order expansion supplies a valid local quadratic metric can enter IEWC through its matched primal/dual pair G_n, G_n^{-1}.

For ordered probability outputs, Appendix F constructs a CDF/Cramér metric and its matched pseu-

doinverse dual on the simplex tangent. This preserves the original motivation for space-dependent output geometry but is not sliced Wasserstein. The diffusion experiment provides exploratory evidence for output-representation flexibility rather than validation of a Wasserstein geometry.

7 IMPROVED ELASTIC WEIGHT CONSOLIDATION

The preceding derivation leaves five implementation choices: global penalty strength, damping, aggregation weights, output geometry, and matrix approximation. For old-task samples $a_1, \ldots, a_N$, IEWC uses

$$M_{\mathrm{IEWC}} = \frac{1}{N}\sum_{n=1}^{N} \bar{h}_n \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau_A}, \qquad g_n = J_n^\top r_n,$$

where $\bar{h}_n \geq 0$ has empirical mean one. The continual-learning penalty is

$$\frac{\lambda}{2}(\theta - \theta_A^\star)^\top M_{\mathrm{IEWC}}(\theta - \theta_A^\star).$$

The canonical geometry is $G_n = I$. We use task-relative damping $\tau_A = \kappa\,\mathrm{median}\{r_i^\top G_i^{-1} r_i : r_i \neq 0\}$ with $\kappa = 10^{-2}$ in the held-out evaluation. This makes κ dimensionless and preserves global loss-unit invariance. If all denominators are zero, the stored importance is zero.

Uniform IEWC sets $\bar{h}_n = 1$. To study weighting without confounding it with normalization, we define a separate *residual-diversity weighting* (RDW) variant. RDW assigns mean-one weights from the component of each unit parameter-gradient direction outside a capped span of earlier directions; Appendix C.1.1 gives the exact streaming rule. Both OPG-RDW and IEF-RDW receive the same unit parameter-gradient directions. Scalar IEF normalization cannot change those directions, so the 2×2 comparison

$$\{\mathrm{OPG}, \mathrm{IEF}\} \times \{\mathrm{uniform}, \mathrm{RDW}\}$$

isolates matrix normalization from sample weighting.

An IEWC instantiation is specified by the five-tuple $(\lambda, \kappa, \Gamma, h, \Pi)$, where λ sets preservation strength, κ sets relative damping, Γ maps an old output to its local metric G_n, h supplies aggregation weights, and Π is the matrix approximation operator (for example, full, diagonal, or low rank). These are genuine design choices rather than values fixed by the derivation. Scalar loss preservation gives OPG-EWC; loss-level-set normal preservation gives uniform IEWC. We call this CLOC-derived mechanism *Improved Elastic Weight Consolidation* while treating its individual axes separately in theory and experiment.

Table 2: Multi-domain results, means over seeds of the selected configurations. Across-seed standard deviations and full-precision means are reported in Appendix Table 8. Higher is better except for MSE and forgetting.

Task	Benchmark	Model	Metric	Seq.EW
Classification	ImageNet-R	ViT-B/16	Acc. ↑	.2896
			Fgt. ↓	.7066
Language	TRACE	Qwen2.5-3B	Score ↑	.5760
			Fgt. ↓	.1139
Forecasting	Weather, ECL, Traffic	PatchTST	MSE ↓	.2733
			Fgt. ↓	.0535
Segmentation	VOC 15–5	ConvNeXtV2-T FPN	mIoU ↑	.0917
			Fgt. ↓	.7135
Diffusion	MNIST	DDPM	MSE ↓	.0676
			Fgt. ↓	.0503

8 EMPIRICAL EVALUATION

We report two evaluations. The multi-domain study measures IEWC's performance across classification, NLP, forecasting, segmentation, and diffusion; because its results also informed method development, they describe selected configurations rather than unbiased effect estimates. The held-out evaluation fixes hyperparameter selection before any test access and estimates paired effects on held-out data. It compares IEF with OPG under matched uniform weighting and diagonal scale, then crosses OPG/IEF with uniform/RDW aggregation. Appendix C specifies data splits, fixed validation grids, repetitions, baselines, and uncertainty calculations.

Unless stated otherwise, all stored matrices are diagonal. EWC, IEWC, and IEWC+RDW denote observed-label OPG, uniform IEF, and residual-diversity-weighted IEF, respectively. RDW is order dependent, so its gains cannot be attributed to normalization alone; Appendix C maps these names to the archived code and gives complete protocols.

8.1 Multi-domain results

Table 2 shows that uniform IEWC achieves higher means than OPG-EWC on ImageNet-R and VOC 15–5, while weighted IEWC achieves the highest TRACE score and the lowest forecasting and DDPM MSE. The held-out evaluation then tests which gains generalize and whether they arise from normalization or weighting.

At the distribution level, Figure 2A–B shows that IEWC's retained accuracy is not concentrated in a single late distribution. The held-out evaluation tests whether this advantage generalizes across fixed class

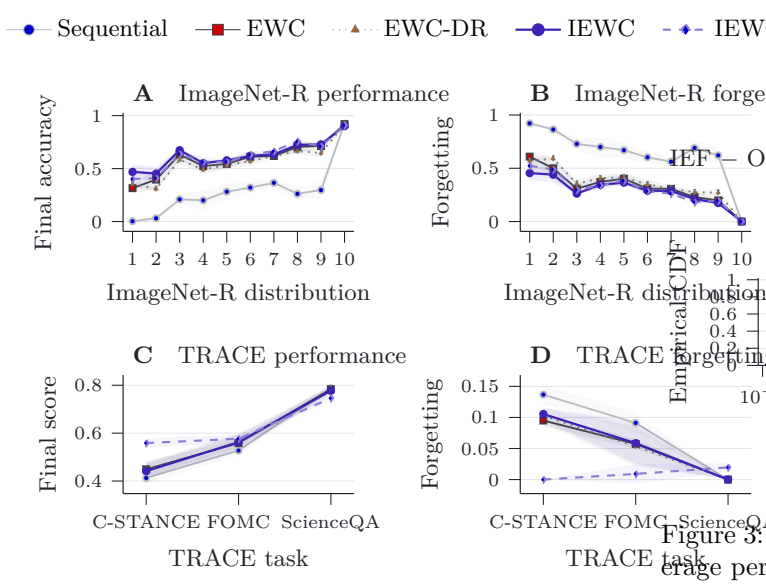


Figure 2: Distribution-level retention in the multi-domain study. **A**–**B**: ImageNet-R final accuracy and forgetting. **C**–**D**: TRACE final score and forgetting for C-STANCE, FOMC, and ScienceQA. Bands show across-seed standard deviations.

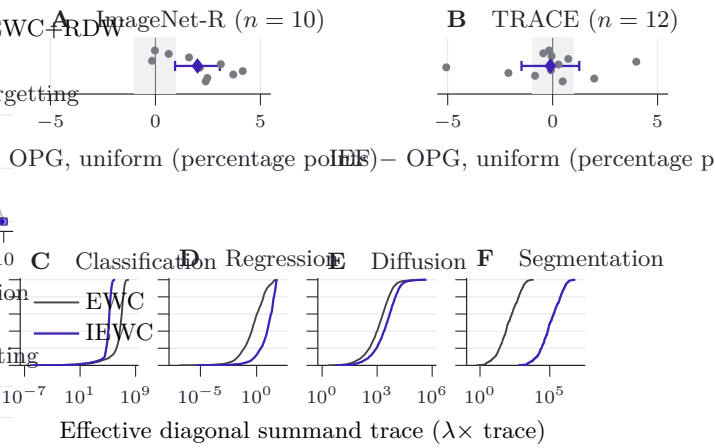


Figure 3: **A**–**B**: Held-out paired differences in final average performance, uniform IEF minus uniform OPG. Gray points are individual paired runs, the diamond is the mean, the bar its 95% Student-t interval, and the shaded band the one-point practical threshold fixed in advance. **C**–**F**: Empirical CDFs of λ times the per-sample diagonal trace for CIFAR-100 classification, controlled regression, MNIST DDPM denoising, and binary segmentation.

orders. The weighted IEWC run likewise has less forgetting on C-STANCE and FOMC in Figure 2C–D. The held-out evaluation crosses the weighting mechanism with both OPG and IEF, and replaces FOMC with NumGLUE-cm to obtain disjoint validation and test splits.

Weighted IEWC retains more target-trajectory structure at the first task and remains close to OPG-EWC later; Appendix Figure 4 provides representative held-out rollouts alongside the aggregate scores.

The DDPM study also uses a row/column cumulative-distribution output representation motivated by sliced Wasserstein distance. The selected run reduced old-output drift to 9.4409 ± 7.2517, compared with 14.1026 ± 12.2220 for Euclidean uniform IEWC, while attaining denoising MSE 0.0301 ± 0.0012 versus 0.0298 ± 0.0014. Because that surrogate did not establish a matched primal/dual metric pair, the result supports output-representation flexibility but not a claim about Wasserstein geometry. Appendix F supplies a correct ordered-CDF/Cramér construction.

Figure 3C–F shows the effective per-sample diagonal summands, λ times the per-sample diagonal trace before averaging, in four representative settings. In classification and regression, OPG summands span more than three orders of magnitude between their 5th and 95th percentiles, while IEWC narrows this range by more than an order of magnitude; in diffusion and segmentation, both spreads are comparable.

8.2 Held-out evaluation

The held-out evaluation selects λ on validation data from fixed five-point grids and evaluates each method once on the test split (Appendix C.2). ImageNet-R uses ten paired class orders; TRACE crosses all six orders of C-STANCE, ScienceQA, and NumGLUE-cm with two seeds. On ImageNet-R, uniform IEWC improves final average accuracy over matched EWC by 2.00 percentage points (95% CI 0.93–3.07; exact sign-flip $p = 0.0078$; Figure 3A), and the IEF advantage persists under RDW aggregation (1.71 points, 0.92–2.51; Holm $p = 0.016$). On TRACE, the difference is -0.11 points (-1.48–1.27; Figure 3B).

9 LIMITATIONS

CLOC is conditional on its preservation specification: it derives a penalty after the preserved functionals and constraint sets have been chosen, but it does not determine which properties are sufficient to preserve old-task competence under future updates. Misspecified or mutually incompatible specifications can therefore protect the wrong behavior even when their local penalties are optimized accurately. How to select or learn

suitable preservation specifications is not addressed.

IEWC, like EWC, compresses an old task into one importance matrix evaluated at $\theta_A^\star$ and keeps it fixed during later training. Before aggregation, however, its normalized parameter-sensitivity direction $a^G(x;\theta_A^\star) = J_{\theta_A^\star}(x)^\top \hat{r}^G(x)$ is input-dependent. Circuit analyses make the same distinction between a model's fixed connections and the input-dependent feature contributions that realize a particular computation (Dunefsky et al., 2024). A single average—especially after diagonalization—can therefore hide sensitivity confined to subsets of inputs or distributed across overlapping pathways, and cannot track changes in the relevant Jacobians as the parameters move away from $\theta_A^\star$. Sample weighting changes which empirical average is stored, but cannot recover this conditional structure after compression. Conditional mixtures or dynamically refreshed importance are natural extensions, at the cost of richer stored summaries or access to old inputs.

Finally, empirical comparisons cover selected datasets, model families, adaptation regimes, and task orders. Validation-based selection and paired held-out evaluation reduce selection bias but cannot establish a domain-independent advantage or deployment robustness. Offline/joint training is only a reference because it violates the continual data-access constraint, and untested replay, distillation, and parameter-isolation methods may be preferable when their storage or architectural assumptions are allowed.

10 CONCLUSION

CLOC is a preservation-first formulation deriving local penalties from selected learned-behavior functionals. Scalar-loss preservation yields an OPG-based regularization that under NLL equals EWC's empirical-Fisher regularization, whereas preserving outputs modulo old-loss level-set tangents yields IEWC and explains its normalization geometrically. CLOC contextualizes damping, geometry, aggregation, and approximation. Mechanism tests, the multi-domain study, and the held-out evaluation show IEWC variants improve EWC in several settings and remain comparable in others.

AI Use Statement

The original IEWC framework and initial manuscript predate the generative-AI assistance disclosed here. Generative AI tools were used to assist with revision planning; theoretical and experimental development; implementation and experiment execution; analysis and interpretation; figure creation; literature search and synthesis; and manuscript drafting and editing. The author reviewed AI-assisted code, experimental outputs, numerical claims, citations, figures, and text against saved artifacts and executable code, and takes responsibility for the final content.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
   (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes. The main theoretical development derives the construction and Appendix F specifies the executable methods.]
   (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes. The well-fitted, output-geometry, notation/boundary, and IEWC sections analyze the construction; Appendix F states storage and importance-estimation costs and Appendix C reports measured time and peak memory.]
   (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes. The supplementary archive contains both the frozen run source and the reproduction source, together with a pinned lockfile.]

2. For any theoretical claim, check if you include:
   (a) Statements of the full set of assumptions of all theoretical results. [Yes. The assumptions appear with the corresponding derivations and in Appendix A.]
   (b) Complete proofs of all theoretical results. [Yes. The results used by the method are proved algebraically in the main text and Appendix A.]

(c) Clear explanations of any assumptions. [Yes. The derivations, notation/boundary section, and limitations section explain their operational consequences.]

3. For all figures and tables that present empirical results, check if you include:
   (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes. The supplementary archive contains source, analysis-complete records, manifests, checksums, and end-to-end instructions; datasets and checkpoints are obtained from their cited sources.]
   (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes. Appendix C records the complete frozen protocol, selection rule, roles, grids, and training configurations.]
   (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes. The empirical-evaluation section and Appendix C define the metrics, repetitions, paired intervals, randomization tests, and dependence checks.]
   (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes. Appendix F reports the single-workstation CPU/GPU environment and the measured resource fields.]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
   (a) Citations of the creator If your work uses existing assets. [Yes.]
   (b) The license information of the assets, if applicable. [Yes. Appendix F states the applicable upstream terms and redistribution boundary.]
   (c) New assets either in the supplemental material or as a URL, if applicable. [Yes. The anonymized supplementary archive contains the new protocol, result records, analyses, and source snapshots.]
   (d) Information about consent from data providers/curators. [Not Applicable. The work uses existing public research benchmarks and does not collect new data from people.]
   (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable to the new artifacts. No benchmark examples are redistributed, and restricted prompt and target fields are removed from archived run records.]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
   (a) The full text of instructions given to participants and screenshots. [Not Applicable.]
   (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable.]
   (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable.]

Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning: Supplementary Materials

A INVARIANCE TO LOSS REPARAMETERIZATION

This appendix treats the undamped case $\tau = 0$.

The output-level feasible set

$$\mathcal{C}_n = \{z : \ell_n(z) = \ell_n(z_n^A)\}$$

depends on the level sets of ℓ_n, not on the numerical labels assigned to those level sets. If we apply a local monotone scalar transformation

$$\tilde{\ell} = \phi(\ell), \qquad \phi'(\ell) > 0,$$

then the output gradient rescales as

$$\tilde{r}_n = \phi'(\ell_n(z_n^A))\, r_n.$$

For OPG, the sample contribution changes by

$$J_n^\top \tilde{r}_n \tilde{r}_n^\top J_n = \phi'(\ell_n(z_n^A))^2 J_n^\top r_n r_n^\top J_n.$$

For IEF, the same scalar factor appears in numerator and denominator:

$$\frac{\tilde{r}_n \tilde{r}_n^\top}{\tilde{r}_n^\top G_n^{-1} \tilde{r}_n} = \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n}.$$

Thus undamped IEF is invariant to local scalar relabelings of the constraint function wherever $r_n \neq 0$. This includes loss rescaling, unit changes in regression losses, reduction conventions, and smooth monotone transformations that preserve the local level-set geometry. OPG's global scale can be absorbed into λ only when the transformation rescales all samples equally; sample-dependent factors change its relative weighting.

Positive shared damping narrows the invariance. A fixed numerical τ does not transform with the loss and therefore changes even under a global unit rescaling. Task-relative damping $\tau_A = \kappa \,\mathrm{median}_{i:s_i>0}\, s_i$ restores invariance to a global task-loss multiplier because s_i and τ_A scale together. It does not generally preserve invariance to sample-dependent monotone transformations. At $r_n = 0$, the implementation assigns a zero contribution rather than claiming an undamped 0/0 identity.

B CONTAMINATION CONTRIBUTIONS AND SPECTRAL STABILITY

The OPG–IEF normalization difference can change which samples dominate a full importance matrix. In Euclidean output geometry, write $s_i = \|r_i\|_2^2$ and $a_i = g_i/\sqrt{s_i}$ for $s_i > 0$. The undamped per-sample matrices are

$$M_{\mathrm{OPG},i} = s_i a_i a_i^\top, \qquad M_{\mathrm{IEF},i} = a_i a_i^\top.$$

Thus OPG couples each sensitivity direction to the squared output-loss-gradient scale, whereas undamped IEF keeps the direction-only outer product. Relative damping multiplies IEF by $s_i/(s_i + \tau_A)$ and therefore retains the near-zero qualification from Section 5.

For intuition, suppose a fraction ε of samples are contaminated, their normalized sensitivity is concentrated near a direction u outside the leading clean subspace, and their scale is $s_i \approx W$. If clean scales are sufficiently concentrated or independent of direction, then

$$\mathbb{E}[M_{\mathrm{OPG}}] \approx (1-\varepsilon)\bar{s}_{\mathrm{clean}}\Sigma_{\mathrm{clean}} + \varepsilon W u u^\top,$$

whereas undamped IEF has no factor W on the contaminating direction. In particular, a sufficiently large εW can move u into the leading OPG eigenspace. This model is illustrative rather than a distribution-free robustness theorem.

We test the mechanism directly in a fitted two-class MLP. For each seed, 512 inputs are drawn from $\mathcal{N}(0, I_2)$ and assigned the clean label $\mathbf{1}\{x_1 + 0.75x_2 < 0\}$; a uniformly selected 0, 5, 10, or 20% of labels is flipped. A 2–32–2 tanh MLP is trained for 250 epochs with Adam, learning rate 0.01, and batch size 64. For each of 20 independent seeds we then retain every per-sample parameter-gradient row and clean/corrupt identity. We decompose the full OPG and task-relative-damped IEF matrices ($\kappa = 0.01$) into clean and corrupt contributions, then compare each total matrix with its clean-contribution reference. Table 3 reports means over seeds.

Table 3: Direct full-matrix contamination diagnostic. Alignment is the absolute inner product of the leading eigenvectors; top-5 overlap is the mean squared singular value between the corresponding eigenspaces. Each reference is the clean contribution from the same fitted model.

Flip rate	Corrupt/clean s_i ratio	Corrupt trace share		Leading-vector alignment		Top-5 overlap	
		OPG	IEF	OPG	IEF	OPG	IEF
0.05	81.26	0.783	0.054	0.426	1.000	0.902	0.996
0.10	36.67	0.789	0.104	0.720	0.999	0.923	0.997
0.20	11.23	0.736	0.202	0.875	0.998	0.930	0.990

Table 4: Paired mean differences with 95% Student-t intervals over 20 seeds. Every interval excludes zero.

IEF minus OPG contrast	5% flips	10% flips	20% flips
Corrupt trace share	-0.729 $[-0.751, -0.706]$	-0.685 $[-0.701, -0.670]$	-0.534 $[-0.549, -0.519]$
Leading-vector alignment	$+0.574$ $[+0.444, +0.704]$	$+0.279$ $[+0.152, +0.406]$	$+0.123$ $[+0.034, +0.213]$
Top-5 overlap	$+0.094$ $[+0.067, +0.122]$	$+0.074$ $[+0.047, +0.101]$	$+0.060$ $[+0.029, +0.091]$

These observations support the proposed contamination mechanism in this controlled construction. They do not establish a universal robustness guarantee or prove that a diagonal continual-learning penalty improves on a real benchmark; those are separate empirical questions.

C EXPERIMENTAL PROTOCOL, PROVENANCE, AND SENSITIVITY

C.1 Held-out evaluation protocol

The evaluation protocol was fixed and hashed before any test result was computed. It specifies the data splits, methods, hyperparameter grids, repetitions, metrics, and statistical analysis. Test data can be loaded only with the matching protocol hash, and every access is logged before examples are loaded. Validation scripts refuse to load test data, and validation runs do not enter the reported intervals.

The primary comparison in each domain is the paired final average performance of uniform IEF minus uniform OPG. The secondary comparisons cross estimator and aggregation:

$$\{\mathrm{OPG}, \mathrm{IEF}\} \times \{\mathrm{uniform}, \mathrm{RDW}\}.$$

All four combinations use the same importance examples, and the RDW variants receive the same unit-normalized parameter-gradient directions. Sampled model-Fisher EWC and MAS (Aljundi et al., 2018) are additional regularization baselines. Sequential and offline/joint training are references. Every nonzero diagonal importance is rescaled to mean one across trainable parameters, and every method uses the displayed $\lambda/2$ convention.

This comparison keeps the same post-task diagonal penalty for every method. SI's trajectory accumulation and LwF's distillation objective are discussed in Section 2 but are not part of it. EWC, IEWC, and IEWC+RDW in the paper correspond to the EF-EWC, IEWC-E/U-IEWC, and IEWC-GSS/GSS-IEWC labels in the archived code.

C.1.1 Residual-diversity weighting rule

For each nonzero parameter gradient, let

$$u_n = \frac{g_n}{\|g_n\|_2}.$$

In a fixed sample order, RDW maintains an orthonormal basis Q of at most K earlier residual directions and assigns

$$h_n = \|(I - Q^\top Q)u_n\|_2^2.$$

When the residual is nonzero and the cap is not reached, its normalized direction is appended to Q; the final h_n are mean-normalized. This streaming rule is motivated by residual-gradient diversity ideas in GSS (Aljundi et al., 2019), but it is an order-dependent aggregation heuristic, not part of the level-set derivation.

Table 5: Data splits and repetitions in the held-out evaluation. FOMC is excluded because its distributed evaluation and test files are exact duplicates.

Domain	Data splits	Validation runs	Test runs
ImageNet-R	Stratified 80%/20% split of the original 24,000-example train set for train/test; the original 6,000-example test set is used only for validation	3 class orders	10 paired class orders
TRACE	Distributed train/eval/test files for C-STANCE, ScienceQA, and NumGLUE-cm; prompt/answer overlaps are zero	3 task orders	6 task orders × 2 optimizer seeds

ImageNet-R uses the fixed λ grid $\{0.01, 0.03, 0.1, 0.3, 1\}$. TRACE uses $10^{-5} \times \{1, 3, 10, 30, 100\}$. Neither grid is extended after validation.

ImageNet-R uses a pretrained ViT-B/16 with LoRA rank 4 and ten 20-class tasks, trained for two epochs per task with AdamW at learning rate $3 \cdot 10^{-4}$ and batch size 64. Importance uses 128 examples per completed task and RDW cap 64. TRACE uses Qwen2.5-3B-Instruct with LoRA rank 8, two epochs, AdamW at $2 \cdot 10^{-4}$, batch size 2 with gradient accumulation 8, 512 training examples per task, and 64 importance examples per completed task. Data, loader, optimizer, task-order, importance-order, and sampled-Fisher-label seeds are separately recorded and paired where appropriate.

All runs use explicit DataLoader generators, reseed each task's training stage independently of the importance computation, and require deterministic PyTorch and CUDA kernels. Before validation, two complete runs with the same configuration were compared in each domain after removing wall-clock metadata; their training losses, evaluation matrices, and, for TRACE, generated example-level predictions matched exactly. These duplicate runs are kept for reproducibility checks but excluded from selection.

For every paired comparison we report the mean and standard deviation of paired differences, a two-sided 95% Student-t interval, the exact two-sided sign-flip p-value when enumerable, the median, and every paired value. These summaries treat task-order/seed pairs as independent; the interval assumes approximately normal paired differences, and the sign-flip test assumes that signs are exchangeable under the null. We interpret results against a practical threshold of 0.01, one percentage point on either zero-to-one final-average metric, fixed in advance rather than relying on a binary significance rule. Holm correction is applied to all five secondary comparisons within a domain. Full five-point validation curves, importance time, total run time, and peak CUDA memory are recorded.

TRACE's twelve runs form a balanced cross of six task permutations and two model/optimizer seeds, so they are not twelve independent order draws. The primary analysis, threshold, and secondary comparisons therefore use all twelve runs, and two dependence checks complement them. First, we average each method over the two seeds within an order and recompute all paired summaries over the six orders. Second, we average over all six orders within each seed and report the two seed-level paired differences without treating a two-unit interval as reliable inference. The balanced design preserves the overall paired mean. Where run-level and order-level uncertainty differ, we base claims about TRACE on the more conservative order-level result. The six orders are the complete set of permutations, not a sample from a larger population of orders.

IEF uses task-relative damping

$$\tau_A = 10^{-2}\,\mathrm{median}\{s_i : s_i > 0\}.$$

Validation-only sensitivity varies the coefficient over $\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, importance counts, RDW caps, and RDW sample order. A separate loss-unit diagnostic multiplies a fitted classification loss by 0.1, 1, and 10. OPG raw scale changes quadratically, fixed numerical damping changes IEF shape, and task-relative damping preserves the IEF matrix to numerical tolerance, matching Section 5.

C.2 Held-out results

Table 6 reports every method in the held-out evaluation, and Table 7 the primary comparison and the Holm-corrected secondary comparisons. Values are fractions; differences in the main text are in percentage points.

Table 6: Held-out final performance and forgetting, mean ± standard deviation over paired runs, with the validation-selected λ.

Domain	Method	Selected λ	Final performance	Forgetting
ImageNet-R	Sequential	–	0.2808 ± 0.0438	0.7018 ± 0.0467
	Offline/joint	–	0.8199 ± 0.0029	0.0000 ± 0.0000
	OPG, uniform	1.0000	0.6416 ± 0.0301	0.2835 ± 0.0337
	IEF, uniform	1.0000	0.6616 ± 0.0168	0.2566 ± 0.0177
	OPG, RDW	1.0000	0.6445 ± 0.0278	0.2803 ± 0.0310
	IEF, RDW	1.0000	0.6617 ± 0.0185	0.2565 ± 0.0196
	Sampled model Fisher	1.0000	0.6477 ± 0.0242	0.2755 ± 0.0259
	MAS	0.0300	0.4602 ± 0.0261	0.5015 ± 0.0273
TRACE	Sequential	–	0.5697 ± 0.0431	0.0758 ± 0.0440
	Offline/joint	–	0.6391 ± 0.0183	0.0000 ± 0.0000
	OPG, uniform	3×10^{-4}	0.5703 ± 0.0395	0.0750 ± 0.0458
	IEF, uniform	3×10^{-4}	0.5692 ± 0.0408	0.0670 ± 0.0471
	OPG, RDW	3×10^{-4}	0.5800 ± 0.0381	0.0685 ± 0.0446
	IEF, RDW	3×10^{-5}	0.5646 ± 0.0395	0.0763 ± 0.0407
	Sampled model Fisher	1×10^{-5}	0.5758 ± 0.0460	0.0759 ± 0.0427
	MAS	1×10^{-4}	0.5650 ± 0.0441	0.0785 ± 0.0433

Table 7: Held-out paired comparisons with 95% Student-t intervals, exact sign-flip p-values, and Holm-corrected p-values for the secondary comparisons.

Domain	Type	Comparison	Mean paired difference	95% CI	Exact p	Holm p
ImageNet-R	Primary	IEF uniform − OPG uniform	0.0200	[0.0093, 0.0307]	0.0078	–
	Secondary	IEF, RDW − OPG, RDW	0.0171	[0.0092, 0.0251]	0.0039	0.0156
	Secondary	OPG, RDW − OPG, uniform	0.0030	[-0.0024, 0.0084]	0.2637	0.5273
	Secondary	IEF, RDW − IEF, uniform	8.5×10^{-5}	[-0.0017, 0.0018]	0.8867	0.8867
	Secondary	Sampled model Fisher − OPG, uniform	0.0061	[-0.0021, 0.0144]	0.1270	0.3809
	Secondary	MAS − OPG, uniform	-0.1814	[-0.1986, -0.1643]	0.0020	0.0098
TRACE	Primary	IEF uniform − OPG uniform	-0.0011	[-0.0148, 0.0127]	0.8589	–
	Secondary	IEF, RDW − OPG, RDW	-0.0155	[-0.0335, 0.0026]	0.0513	0.2563
	Secondary	OPG, RDW − OPG, uniform	0.0097	[-0.0042, 0.0237]	0.1914	0.7656
	Secondary	IEF, RDW − IEF, uniform	-0.0047	[-0.0136, 0.0043]	0.2759	0.8276
	Secondary	Sampled model Fisher − OPG, uniform	0.0055	[-0.0063, 0.0172]	0.3887	0.8276
	Secondary	MAS − OPG, uniform	-0.0053	[-0.0156, 0.0050]	0.3013	0.8276

C.3 Multi-domain protocol

Table 9 records the configurations behind Table 2. In the multi-domain study, the reported results also informed grid placement, grid extension, and method selection, so they are not unbiased effect estimates. Its implementation omits the factor 1/2 of the displayed penalty, so its λ values are not directly comparable with those of the held-out evaluation.

Table 8: Complete multi-domain results corresponding to Table 2, reported as mean (across-seed standard deviation). Higher is better for accuracy, score, and mIoU; lower is better for MSE and forgetting.

Task	Metric	Seq.	EWC (OPG)	EWC-DR	IEWC	IEWC+RDW
Classification	Final avg. acc. ↑	0.2896(.0405)	0.5981(.0337)	0.5687(.0263)	**0.6332(.0232)**	0.6250(.0263)
	Avg. forgetting ↓	0.7066(.0466)	0.3595(.0378)	0.3933(.0286)	**0.3142(.0250)**	0.3285(.0278)
NLP	Final avg. score ↑	0.5760(.0027)	0.5972(.0020)	0.5972(.0059)	0.5938(.0034)	**0.6272(.0080)**
	Avg. forgetting ↓	0.1139(.0111)	0.0762(.0034)	0.0781(.0109)	0.0820(.0122)	**0.0046(.0079)**
Forecasting	Final avg. MSE ↓	0.2733(.0065)	0.2388(.0021)	–	0.2418(.0038)	**0.2383(.0007)**
	MSE forgetting ↓	0.0535(.0081)	**0.0140(.0020)**	–	0.0191(.0047)	0.0147(.0021)
Segmentation	Final all mIoU ↑	0.0917(.0061)	0.0946(.0040)	0.3080(.0307)	**0.4678(.0568)**	0.4274(.0574)
	Base forgetting ↓	0.7135(.0114)	0.7131(.0166)	0.4471(.0447)	**0.2418(.0965)**	0.2925(.0839)
Diffusion	Final denoise MSE ↓	0.0676(.0029)	0.0296(.0013)	–	0.0298(.0014)	**0.0293(.0013)**
	MSE forgetting ↓	0.0503(.0002)	**-0.0006(.0020)**	–	0.0001(.0024)	-0.0003(.0020)

Table 9: Configurations for the multi-domain study. "DR" denotes EWC-DR.

Task type	Dataset, sequence, and model	Selected values
Classification	ImageNet-R; ten 20-class distributions; pretrained ViT-B/16 with LoRA rank 4.	OPG: $\lambda = 333.333$; DR: $\lambda = 3$; U-IEWC: $\lambda = 33.333, \tau = 10^{-4}$; weighted IEWC: $\lambda = 11.111, \tau = 10^{-2}$.
NLP	TRACE, C-STANCE → FOMC → ScienceQA; pretrained Qwen2.5-3B with LoRA rank 8.	OPG: $\lambda = 1$; DR: $\lambda = 1$; U-IEWC: $\lambda = 0.111, \tau = 10^{-2}$; weighted IEWC: $\lambda = 59049, \tau = 10^{-2}$.
Forecasting	Weather → ECL → Traffic; PatchTST trained from scratch.	OPG: $\lambda = 729$; U-IEWC: $\lambda = 9, \tau = 10^{-2}$; weighted IEWC: $\lambda = 9, \tau = 10^{-2}$.
Segmentation	VOC 15–5, base then novel classes; pretrained ConvNeXtV2-tiny FPN with full fine-tuning.	OPG: $\lambda = 0.111$; DR: $\lambda = 243$; U-IEWC: $\lambda = 270, \tau = 10^{-3}$; weighted IEWC: $\lambda = 2430, \tau = 10^{-2}$.
Diffusion	MNIST DDPM, labels 0/1 → 2/3; conditional DDPM trained from scratch.	OPG: $\lambda = 9 \cdot 10^5$; U-IEWC: $\lambda = 675, \tau = 10^{-3}$; weighted IEWC: $\lambda = 225, \tau = 10^{-3}$; output ablation: $\lambda = 10^6, \tau = 10^{-3}$.

The multi-domain search used a five-point multiplicative grid around a task-specific midpoint and extended the grid up to three times when an endpoint was best. Its results therefore describe the performance of the selected configurations rather than unbiased effect estimates. The held-out evaluation instead fixes all five grid points before validation, never extends the grid, and accesses the test split only after the selected hyperparameters are saved.

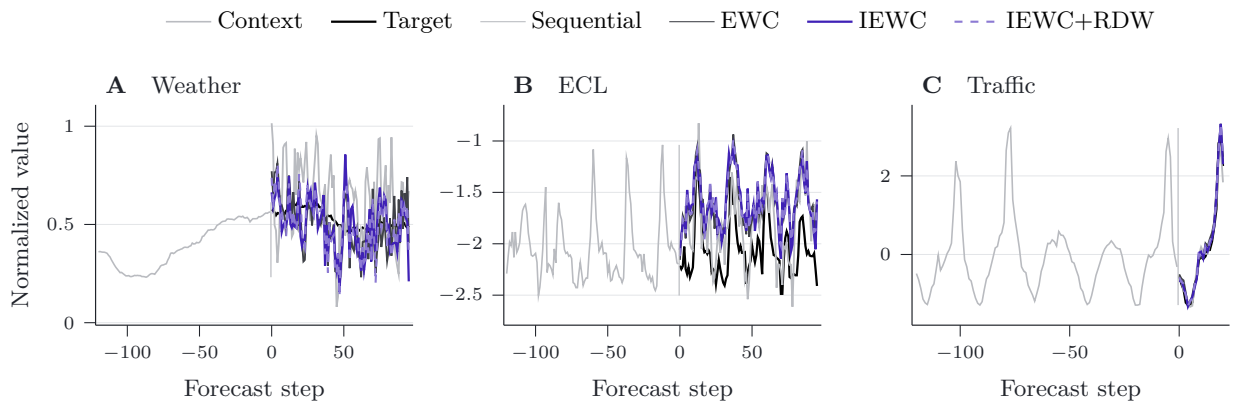


Figure 4: Held-out forecasting rollouts after the continual sequence. Gray: observed context; black: target horizon; colored: method predictions.

The archive also contains the complete log of development runs behind the five reported settings, with 635 per-λ outcomes: 148 vision-classification, 161 forecasting/regression, 47 DDPM, 55 segmentation, 110 encoder-language, and 114 TRACE decoder-only runs. The log includes pilots, smoke checks, tuning sweeps, selected-seed repetitions, and negative or mixed configurations across CIFAR-100 and ImageNet-R; ETT, M4, and long-horizon forecasting; synthetic and GLUE-family language sequences; three Qwen scales on TRACE; VOC 15–5; and MNIST DDPM. It also covers sequential, OPG-EWC, EWC-DR, uniform IEWC, residual-diversity, and FROMP-derived variants where implemented. Table 2 thus reports performance in the strongest representative multi-domain settings, the log exposes the full development record, and the held-out evaluation provides comparisons free of selection bias.

C.4 Damping sensitivity

Figure 5 combines two complementary damping diagnostics on a common higher-is-better change scale. Panel A shows a fixed-τ sweep on a CIFAR-100/FACIL prefix as changes from $\tau = 10^{-2}$: the two curves are final accuracy and negative forgetting, so values below zero indicate worse performance on either measure. Panel B reports a relative-κ sweep, fixed in advance, at the selected λ on one validation repetition per domain, relative to the default $\kappa = 10^{-2}$ of the held-out evaluation. The default is not uniformly best in these single runs: $\kappa = 1$ changes validation performance by $+0.0263$ on ImageNet-R and $+0.0007$ on TRACE. The former exceeds the one-point practical threshold, while the latter is negligible. These results motivate reporting and tuning damping, rather than changing the fixed setting of the held-out evaluation after the fact. Fixed τ has loss-dependent units; relative κ is dimensionless and preserves global loss-unit invariance.

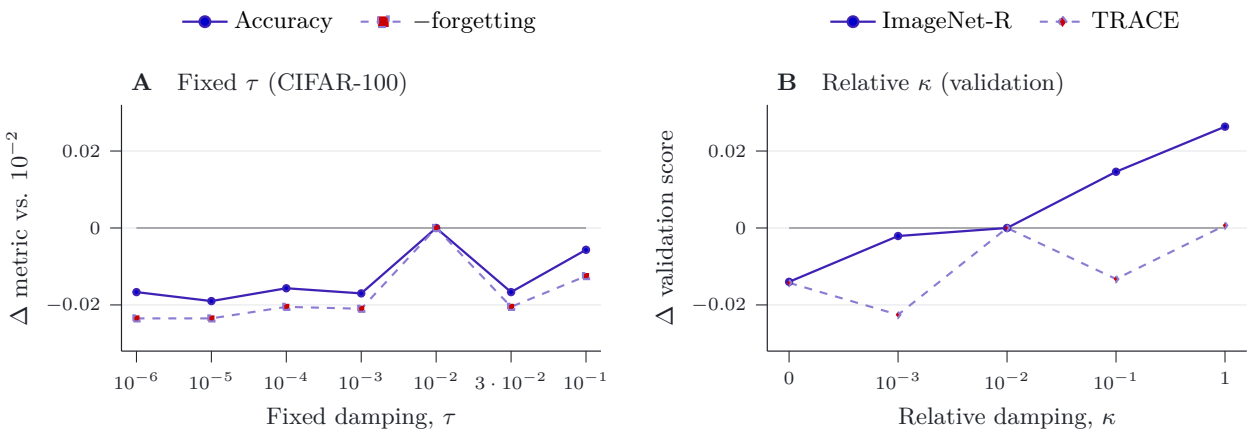


Figure 5: Damping sensitivity. **A**: Fixed-τ sweep on a CIFAR-100 prefix, expressed relative to $\tau = 10^{-2}$; for comparability, forgetting is sign-reversed so higher is better. **B**: Validation-only task-relative damping at fixed selected λ, expressed relative to the default $\kappa = 10^{-2}$. Panel B uses one validation repetition per domain and was not used for selection.

An additional EWC-DR check tested $\lambda \in \{100, 300, 1000, 3000, 10000\}$ on the same CIFAR-100 prefix and selected 100; it is not part of the ImageNet-R held-out selection.

D LABEL-NOISE STRESS TEST

The CIFAR-100 label-noise experiment provides a downstream stress test alongside the direct matrix analysis in Appendix B. Its results were available during method development, so the table reports the performance of selected configurations rather than unbiased effect estimates. Uniform training-label flips create atypical high-loss samples, and the clean task-aware test set measures downstream retention. Across this stress test, IEWC loses less accuracy than observed-label OPG-EWC as the flip rate increases.

Table 10: Final average task-aware accuracy under uniform training-label flips, reported as mean ± across-seed standard deviation.

Method	$p = 0$	$p = 0.1$	$p = 0.25$
OPG-EWC	0.5617 ± 0.0062	0.5132 ± 0.0112	0.2773 ± 0.0320
EWC-DR	0.5932 ± 0.0296	0.5310 ± 0.0226	0.4598 ± 0.0095
IEWC	0.5978 ± 0.0260	0.5642 ± 0.0215	0.5322 ± 0.0239

The multi-domain study also compared sorted diagonal entries on a CIFAR-100 prefix. That diagnostic does not identify clean and corrupt sample contributions and cannot establish the spectral mechanism by itself; Appendix B supplies the direct per-sample full-matrix test. The diagonal summary remains useful as a complementary downstream view. It uses all 512 exported entries per matrix and reports the stored-tail mass ratio, normalized stored-tail profile change, and log-scale RMSE of the sorted entries.

Table 11: Sorted-diagonal diagnostics under label noise.

Method	Noise	Entries	Tail mass ratio	Profile L1	Log-scale RMSE
OPG-EWC	0.1	512	0.0288	0.2610	1.4660
OPG-EWC	0.25	512	0.0244	0.2737	1.5337
IEWC	0.1	512	1.0833	0.0350	0.0350
IEWC	0.25	512	1.1003	0.0154	0.0415

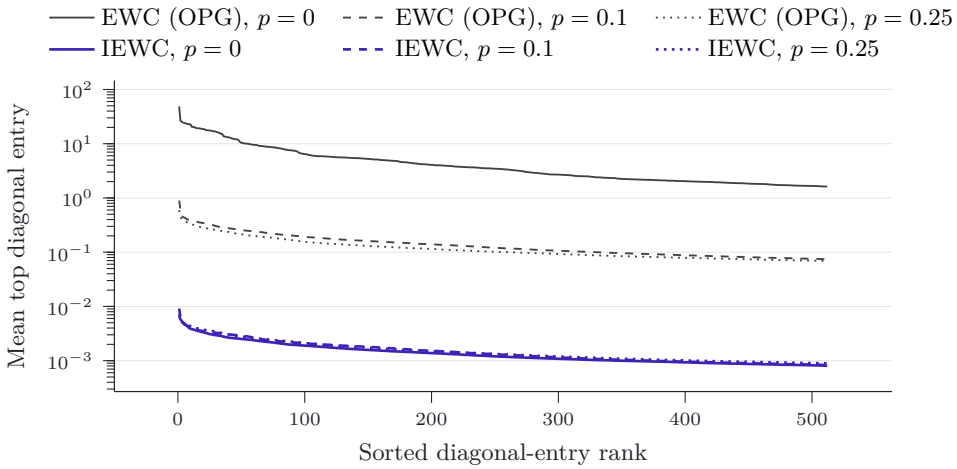


Figure 6: Label-noise diagonal diagnostic. The plot visualizes the sorted high-importance coordinate tail and complements the direct clean/corrupt contribution analysis in Appendix B.

E REPRODUCIBILITY AND ARTIFACT CONTENTS

The supplementary archive supports two reproduction paths. First, all paper tables, intervals, and figures can be regenerated on CPU from the saved result records, without downloading benchmark data or retraining models. Second, the model results can be rerun from the archived source after obtaining the datasets and model checkpoints under their original terms. The archive contains the source snapshot used for all runs with its Git history, a later snapshot used for reproduction, the manuscript source, the evaluation protocol, the selected hyperparameters, validation and test records, sensitivity records, mechanism diagnostics, reporting scripts, a pinned dependency lockfile, and SHA-256 checksums for every file. A verification script checks file counts, paths, hashes, and source commits, and confirms that no restricted dataset examples are included.

The empirical scripts use one `IEWCConfig` interface across the diagonal and low-rank implementations. The same interface records output geometry, damping, aggregation weights, and matrix approximation choices, and is used by both the experiment and reproduction entry points.

Environment and execution. All held-out evaluation records come from source commit `5e170ea7d84d7094dd1734ee1516b9fba2133a63`. The environment is reconstructed with `uv sync -frozen -dev`, and the source passes 75 CPU tests. Runs enable deterministic PyTorch algorithms, disable cuDNN benchmarking and TF32, and separately record model, data-loader, importance-order, and model-Fisher sampling seeds. Each task reseeds training before constructing its data loader, so method-specific importance computation cannot change a later task's minibatch order. Hyperparameter selection rejects missing or duplicate runs and runs from mismatched commits, and test examples are loaded only after the selected hyperparameters have been saved, with every test access logged.

The held-out evaluation comprises 192 validation runs, 176 test runs, and 48 validation-only sensitivity runs. ImageNet-R uses ten paired class orders. TRACE uses all six permutations of its three tasks, crossed with two optimizer seeds; the analysis additionally averages within task order so that the twelve runs are not treated as twelve independent order draws. The contamination experiment uses twenty independent seeds at each of four corruption levels. Every planned run is reported regardless of the direction of its effect. The analysis reports paired differences, Student-t intervals, exact sign-flip tests where enumerable, a one-point practical threshold fixed in advance, and all Holm-corrected secondary comparisons.

Data and model provenance. The archive redistributes no benchmark images, language examples, or pretrained weights. ImageNet-R is obtained from its authors; split definitions are pinned to Mammoth commit `e75a491c69fd729edeb01431afb753d9157d9a81`. The splits contain 19,206 training, 6,000 validation, and 4,794 test images across 200 classes. The ViT-B/16 checkpoint is pinned to Hugging Face revision `063c6c38a5d8510b2e57df480445e94b231dad2c`. TRACE is obtained from its authors under the Apache-2.0

repository license. Its distributed train, evaluation, and test files define the three splits for C-STANCE, ScienceQA, and NumGLUE-cm; their hashes are recorded, and the splits share no prompts or prompt–answer pairs. FOMC is excluded because its distributed evaluation and test files are identical. Qwen2.5-3B-Instruct is pinned to revision `aa8e72537993ba99e69dfaafa59ed015b17504d1` and remains subject to its model terms. Mammoth is MIT licensed, and the pinned `timm` checkpoint declares Apache-2.0. The IEWC source is supplied for review; the submission does not override third-party terms or redistribute their assets.

Compute and complexity. The held-out experiments ran on one workstation with an Intel Core i9-13900K CPU and one NVIDIA GeForce RTX 5090 with 32 GiB of memory. Every run records total wall time, and every importance phase records wall time and peak CUDA allocation. With d trainable parameters and N importance samples, diagonal uniform OPG and IEWC store $O(d)$ values and require N per-sample gradients. RDW maintains at most K unit residual directions on CPU, using $O(Kd)$ additional storage and $O(NKd)$ inner-product work before the same diagonal aggregation. The archive includes a table of the measured cost of every method in both domains.

Regeneration checks. The reproduction guide gives exact commands to rebuild the held-out analysis and all generated tables and figures in a fresh directory. Their checksums are identical across rebuilds once volatile PDF timestamps are removed. A clean-checkout test reconstructs the environment, runs the test suite, regenerates the analysis and paper figures, compiles the manuscript, and compares the resulting checksums. For the multi-domain study, the archive includes the complete log of 635 development runs with per-file checksums; records that may contain third-party prompts or targets are not redistributed.

F MATCHED ORDERED-CDF OUTPUT GEOMETRY

Consider probability vectors on an ordered grid with m bins. Let

$$P = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$$

project onto the probability-simplex tangent, let $C \in \mathbb{R}^{(m-1)\times m}$ compute cumulative sums through the penultimate bin, and let Δx be the grid spacing. Define

$$G_{\mathrm{CDF}} = \Delta x\, PC^\top CP.$$

For a zero-sum probability perturbation Δz,

$$\Delta z^\top G_{\mathrm{CDF}} \Delta z = \Delta x\, \|C\Delta z\|_2^2,$$

which is the squared L^2 distance between the corresponding one-dimensional cumulative distribution functions. Because G_{CDF} is singular in the ambient space, its use is restricted to the probability-simplex tangent. The matched dual form for a loss covector r is

$$(Pr)^\top G_{\mathrm{CDF}}^{+}(Pr),$$

where G_{CDF}^{+} is the Moore–Penrose pseudoinverse. The implementation tests both the primal identity and the matched dual construction numerically.

This is an ordered-CDF or Cramér metric, not sliced Wasserstein. It supplies one explicit space-dependent geometry compatible with IEWC's matched primal/dual construction. The diffusion output-representation experiment does not use this construction and is therefore treated only as exploratory evidence that alternative output representations can affect the penalty, not as empirical validation of Wasserstein geometry.

G PROOF OF THE WELL-FITTED SCALING PROPOSITION

For $s_n > 0$, define

$$\hat{r}_n^{,G} = \frac{r_n}{\sqrt{s_n}}, \qquad b_n^{,G} = \|J_n^\top \hat{r}_n^{,G}\|_2^2.$$

Direct substitution gives

$$S_{n,0}^{\mathrm{IEF},G} = J_n^\top \hat{r}_n^{,G} (\hat{r}_n^{,G})^\top J_n, \qquad S_n^{\mathrm{OPG}} = s_n S_{n,0}^{\mathrm{IEF},G},$$

and therefore

$$\|S_{n,0}^{\mathrm{IEF},G}\|_2 = b_n^G, \qquad \|S_n^{\mathrm{OPG}}\|_2 = s_n b_n^G.$$

Since $G_n \preceq M_G I$, we have

$$\|\hat{r}_n^G\|_2^2 = \frac{\|r_n\|_2^2}{r_n^\top G_n^{-1} r_n} \le M_G.$$

The bounded-Jacobian assumption consequently gives $b_n^G \le C_J^2 M_G$, which proves the two upper bounds. The nondegeneracy condition $\|J_n^\top \hat{r}_n^G\|_2 \ge c_J$ gives the corresponding lower bounds.

For Euclidean softmax cross-entropy with correct-class probability $p_{n,c} = 1-\varepsilon$, the non-target probabilities sum to ε, and hence

$$\varepsilon^2 \le \|p_n - y_n\|_2^2 = \varepsilon^2 + \sum_{k\neq c} p_{n,k}^2 \le 2\varepsilon^2.$$

Thus $s_n \in \Theta(\varepsilon^2)$, which gives the stated rates under the same nondegeneracy condition.

For positive damping, multiplying and dividing by s_n yields

$$S_{n,\tau}^{\mathrm{IEF},G} = \frac{s_n}{s_n+\tau} S_{n,0}^{\mathrm{IEF},G}.$$

With fixed $\tau > 0$, bounded J_n gives $\|J_n^\top r_n\|_2^2/(s_n+\tau) \to 0$ as $r_n \to 0$. Finally, under a global positive loss rescaling by c, both r_n and $g_n = J_n^\top r_n$ scale by c, while s_n and the task-relative τ_A scale by c^2. Their quotient is therefore unchanged.

H ADDITIONAL OUTPUT-SPACE DERIVATIONS

H.1 Matched-metric hyperplane projection

For the tangent hyperplane $T = \{t : r^\top t = 0\}$, write the normal residual as $e = \Delta z - t$ with $t \in T$. The projection solves

$$\min_e e^\top G e \quad \text{subject to} \quad r^\top e = r^\top \Delta z.$$

The Lagrange equations give

$$e^\star = \frac{r^\top \Delta z}{r^\top G^{-1} r} G^{-1} r,$$

whose squared G-norm is

$$\|e^\star\|_G^2 = \frac{(r^\top \Delta z)^2}{r^\top G^{-1} r}.$$

Hence G measures primal output displacement and G^{-1} acts on the loss covector.

H.2 Scalar least-squares reduction

Consider scalar-output least-squares regression in Euclidean geometry,

$$\ell_n(z) = \frac{1}{2}(z - y_n)^2.$$

Then $r_n = z_n^A - y_n$ is scalar and, for $r_n \neq 0$, the undamped normalized factor is $r_n^2/r_n^2 = 1$. Each IEF summand therefore reduces to $J_n^\top J_n$, and averaging gives

$$M_{\mathrm{IEF}} = \frac{1}{N}\sum_{n=1}^N J_n^\top J_n.$$

This is exactly the Gauss–Newton or output-Jacobian form for squared-error regression (Schraudolph, 2002; Martens, 2010, 2020), as also made explicit by Wu et al. (2024).

I NOTATION

Table 12: Notation and distinctions used throughout the paper.

Symbol	Meaning
$g_i = J_i^\top r_i$	Parameter gradient of the selected scalar functional; IEWC uses the old-sample loss.
M_{OPG}	Mean $g_i g_i^\top$ from scalar-loss preservation; only the likelihood-score case is the observed-label empirical-Fisher algebra.
F_{model}	Fisher information averaging score outer products over labels sampled from the frozen predictive distribution.
$s_i = r_i^\top G_i^{-1} r_i$	Squared dual norm of the output-loss gradient under the matched output metric.
M_{IEF}	Level-set-normalized mean $g_i g_i^\top/(s_i + \tau_A)$; the undamped Euclidean case is iEF.
$\rho_\tau(s)$	Damping attenuation $s/(s+\tau)$ relative to the undamped summand.
κ	Dimensionless relative-damping coefficient in $\tau_A = \kappa\,\mathrm{median}_{s_i>0}\, s_i$.
$\bar{h}_i$	Optional mean-one aggregation weight; uniform weighting and RDW are evaluated separately.
(Γ, h, Π)	Output-metric rule, aggregation-weight rule, and matrix approximation operator.